



University
of Exeter

Lecture 17 – Spiking Neural Networks

ECM3412/ECMM409 – Nature-Inspired Computation

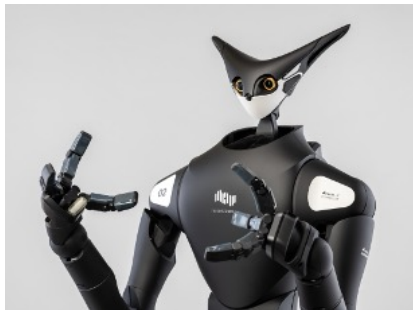
Dr David Walker

Department of Computer Science
D.J.Walker2@exeter.ac.uk

Autumn 2025

Robots

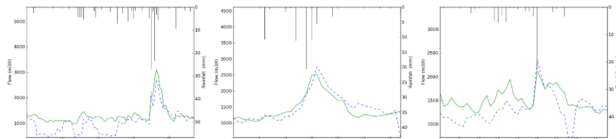
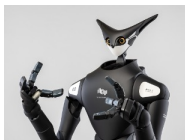
- Perceive their environment using high-dimensional data obtained from sensors.
- Undertake energy efficient and low-latency processing of redundant and sparse information.
- Self-learn to adapt to dynamic and uncertain environments.

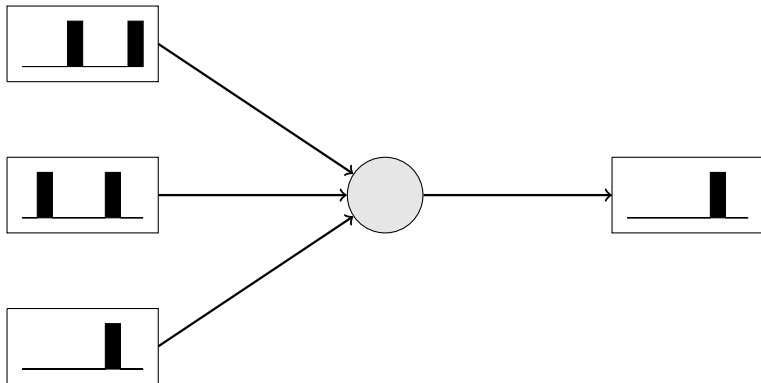


Conventional artificial neural networks cannot do this

- Time consuming and computationally expensive to train, high latency.
- Energy hungry

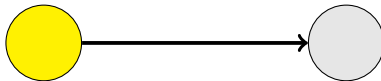
- Mimic the brain's neural activity by transmitting information as **discrete spikes** over time – useful for sequential/time-varying data
- Much more “**biologically plausible**” than traditional ANNs
- Process data sparsely – neurons **only activated when necessary** enabling low-power computing
- Applications: Robotics, speech recognition, sensory processing, and other domains requiring dynamic, temporal pattern recognition.
- Information is encoded in spike rates (rate coding), timing (temporal coding), or distributed across neuron populations (population coding)



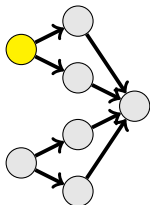


- Sequence of spikes is called a **spike train**
- Each neuron receives and emits spikes

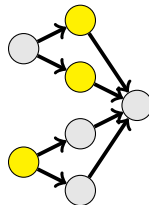
- Neuron charge accumulates **over time** either from input signals or from signals received from other neurons
- When the neuron's charge reaches a **threshold** the neuron fires and emits a signal along each of its outgoing connections
- Neurons have a notion of **leakage** such that below-threshold charge dissipates over time



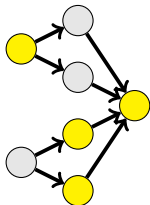
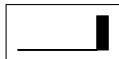
$t = 1$



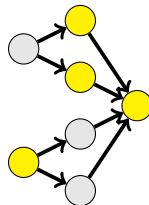
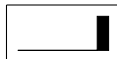
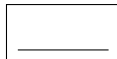
$t = 2$



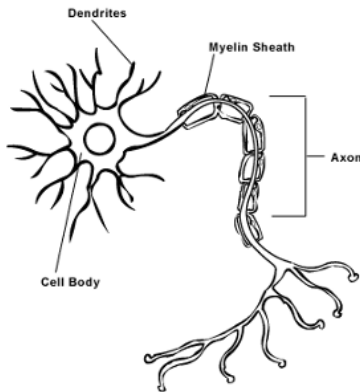
$t = 3$



$t = 4$



- Synapses can adapt their strength based on activity patterns
- The **pre-synaptic** neuron transmits a signal that is received by the **post-synaptic** neuron
- **Potentialiation**: if the pre-synaptic neuron fires before the post-synaptic neuron the connection is strengthened
- **Depression**: if the post-synaptic neuron fires before the pre-synaptic neuron the connection is weakened
- Supports **unsupervised learning**



Adapt the weights – start at output layer and work backwards

$$\mathbf{w}_{ij}^{(t+1)} = \mathbf{w}_{ij}^{(t)} + \eta \delta_{pj} \mathbf{o}_{pj}$$

- $\mathbf{w}_{ij}^{(t+1)}$: weight at time $t + 1$ (the next iteration)
- $\mathbf{w}_{ij}^{(t)}$: weight at time t (the current iteration)
- η : learning rate
- δ_{pj} : error for pattern p on node j
- $\mathbf{o}_{pj}$: output signal for p on j (t_{pj} is the corresponding target)

Computing the error

- Output units: $\delta_{pj} = z o_{pj}(1 - o_{pj})(t_{pj} - o_{pj})$
- Hidden units: $\delta_{pj} = z o_{pj}(1 - o_{pj})(\sum_k \delta_{pk} w_{pk})$
- where z is the sigmoid derivative, $f'(a) = f(a)(1 - f(a))$
- and $(\sum_k \delta_{pk} w_{pk})$ is the weighted sum of error of k units above



The derivative of a neuron's **spiking function** is the **Dirac delta function**, characterised by:

$$\delta(x) = 0, \quad \text{for all } x \neq 0$$

and

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

No gradient information – cannot use backpropagation

Replace the spiking function with a surrogate – e.g. **sigmoid function**

$$f(a) = \frac{1}{1 + e^{-a}},$$

where

$$f'(a) = f(a)(1 - f(a)),$$

which produces a bell-shaped curve centred at $a = 0$ (similar to a Gaussian) – can introduce a scaling factor k , such that

$$f_k(a) = \frac{1}{1 + e^{-kx}}$$

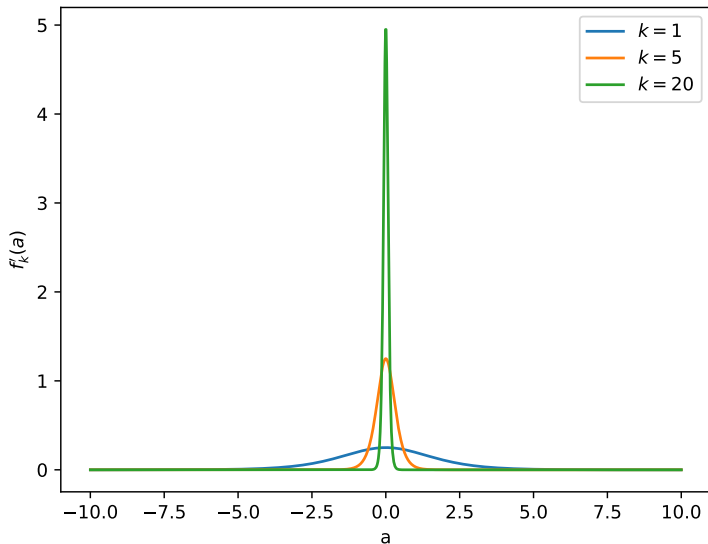
and

$$f'(a) = kf_k(a)(1 - f_k(a)),$$

where

$$\lim_{k \rightarrow \infty} f'_k(a) = \delta(a)$$

Surrogate spiking function - k





Optimise **network structure** and parameters such as **weights**, **delays**, **thresholds**...

Neural architecture search

- Often formulated as a **bi-level** optimisation problem (e.g. optimise the structure and optimise the weights)
- Periodically switch between optimising the structure (the **upper** problem) and the weights (the **lower** problem)

Challenges

- ① Cost of optimisation – more expensive to optimise SNNs than ANNs and we're combining with a highly iterative process (the EA)
- ② Consider wider co-evolution (e.g. multiple aspects of the SNN optimised together)



Spiking neural networks

- Biologically plausible
- Energy efficient
- Cannot be easily optimised with back propagation

Next time

- Neuromorphic computing